F403 — U-3.4 (phase transitions = eigenvalue crossings, Elie's pointer) via the Wallach/spectral machinery: the MUON sits at the spectral phase boundary — the Wallach continuous-series edge ν=a/2=N_c/2=3/2 — between the continuous holomorphic series (gen-1 e/up, ν=5/2>3/2) and the discrete Wallach points (gen-2 μ/charm AT the edge ν=3/2; gen-3 τ/top at the bottom ν=0, minimal rep). Eigenvalue crossings = the formal-degree d(ν) zeros (ν=5/2,1,2,3,4; Plancherel measure vanishes → discrete-series rep hits continuous threshold). The Wallach edge ν=(r−1)a/2=a/2=N_c/2 is the spectral threshold; the muon is the threshold/edge generation (last discrete unitary point before the continuum), explaining its special status. TIER: muon-at-the-edge rigorous (F395 Wallach, a=N_c G4-verified); 'phase transition' = physics framing of the spectral threshold (discrete↔continuous rep character). Target-innocent. Addresses U-3.4. Count 9/26.

Lyra (Claude Opus 4.8)

2026-06-29 Monday (date-verified)

F403 — U-3.4: the muon at the Wallach spectral phase boundary

Elie's U-3.4 (phase transitions = eigenvalue crossings) via the Wallach/eigenvalue machinery.

Eigenvalue crossings = the formal-degree d(ν) zeros (ν=5/2,1,2,3,4; the Plancherel measure vanishes → the discrete-series rep hits the continuous threshold).

Wallach phases: discrete points {0, a/2}={0, 3/2}; continuous (3/2,∞); edge (threshold) at ν=(r−1)a/2 = a/2 = N_c/2 = 3/2. The generations straddle it: gen-1 e/up (ν=5/2) in the continuous series; gen-2 μ/charm (ν=3/2) AT THE EDGE; gen-3 τ/top (ν=0) at the bottom Wallach point (minimal rep).

→ The muon sits at the spectral phase boundary (the Wallach continuous-series edge) — the threshold generation, explaining its special status (the last discrete unitary point before the continuum). Rigorous (F395, a=N_c); the "phase transition" is the physics framing of the spectral threshold. Target-innocent. Count 9/26.